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Continuing Education Activity

Zolpidem is a non-benzodiazepine receptor modulator that is primarily indicated for the short-term alleviation of insomnia, particularly in individuals with sleep initiation challenges. This activity discusses zolpidem and its effectiveness in alleviating sleep latency duration and curtailing nocturnal awakenings, especially in cases of transient insomnia. Zolpidem is also used as an adjunct in augmenting sleep quality for those confronting chronic insomnia, owing to its minor muscle relaxant properties. Zolpidem's pharmacological attributes, optimal dosing considerations, adverse effects, and pertinent contraindications will also be discussed. The activity emphasizes interprofessional collaboration for the adept management of insomnia.

Objectives:

- Identify the indications for zolpidem.
- Evaluate the mechanism of action of zolpidem.
- Assess the adverse effects of zolpidem.
- Implement effective collaboration and communication among interprofessional team members to improve outcomes and treatment efficacy for patients who might benefit from zolpidem therapy for insomnia.

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Indications

FDA-Approved Indications

Zolpidem tartrate is a non-benzodiazepine receptor modulator primarily used in the FDA-approved short-term treatment of insomnia aimed at patients with difficulty falling asleep. Zolpidem belongs to the control substance class C IV medicines and is regulated by the Drug Enforcement Agency. This drug improves measures of sleep latency and sleep duration and reduces the number of awakenings in patients with transient insomnia. Zolpidem also improves sleep quality in patients with chronic insomnia and can act as a minor muscle relaxant. The American Society of Sleep Medicine (AASM) recommends zolpidem for sleep onset and sleep maintenance insomnia in adults. In case the patient continues to have persistent insomnia after 7 to 10 days of zolpidem treatment, it is recommended to conduct a thorough evaluation to identify any underlying medical or psychiatric conditions.

Off-Label Uses

Research has also shown that zolpidem is rapid and effective in restoring brain function in patients who are in a vegetative state after brain injury. The drug has the potential to completely or partially reverse the abnormal metabolism of damaged brain cells. Patients typically recover if the injury is in non-brain stem areas.

Mechanism of Action

Zolpidem is a non-benzodiazepine hypnotic agent that works as a gamma-aminobutyric acid (GABA), a type of receptor chloride channel modulator/agonist that increases GABA inhibitory effects leading to sedation. The GABAa receptor, also called GABA-BZ, is found in the sensorimotor cortical regions, globus pallidus, inferior colliculus, pons, ventral thalamic complex, olfactory bulb, cerebellum, and substantial in the brain. The drug upregulates these receptors, allowing for the sedative effects leading to the preservation of deep sleep.

Unlike benzodiazepines, which non-selectively bind to and activate all benzodiazepine (BZ) receptor subtypes, zolpidem binds the BZ1 receptor preferentially with a high-affinity ratio of the alpha1/alpha5 subunits. The selective binding of zolpidem on the BZ1 receptor may explain the relative absence of myorelaxant and anticonvulsant effects. Zolpidem is generally not recommended as a first-line treatment due to its misuse/abuse potential.. Drugs like controlled-release melatonin and doxepin may be used as the first-line therapy in addition to proper sleep hygiene and cognitive behavioral therapy for patients with insomnia.

Pharmacokinetics

Absorption: Zolpidem is swiftly absorbed from the gastrointestinal tract, whether in immediate-release or controlled-release forms. The immediate-release version achieves peak concentration (Tmax) in approximately 1.6 hours. When taken with food, immediate-release zolpidem exhibits a 15% reduction in mean AUC and a 25% decrease in Cmax. The Tmax is extended by 60% (from 1.4 to 2.2 hours), while the half-life remains unaltered. These findings suggest that, for faster sleep onset, zolpidem tartrate tablets should not be ingested with or immediately after a meal. The controlled-release formulation maintains plasma levels for over 3 hours post-administration. When administered as a single 12.5 mg dose, these tablets have a median Tmax of 1.5 hours. In a food-effect study involving the controlled-release tablets, food consumption causes a 23% reduction in mean AUC, along with prolongation of median Tmax from 2 to 4 hours. The sublingual formulation of zolpidem tartrate disintegrates within the sublingual cavity after administration. On average, it is rapidly absorbed in both genders, with a mean Tmax ranging from about 35 minutes to 75 minutes. Zolpidem tartrate is cleared from the body at a slower rate in women than in men. Female subjects exhibit about 45% higher Cmax and AUC parameters than male subjects when administered at the same dose. Therefore, adult women require a lower initial dose of 5 mg, while adult men can take 5 mg or 10 mg.

Distribution: Total protein binding for both forms of zolpidem is high (92.5 ± 0.1%).

Metabolism: Zolpidem undergoes primary metabolism through various enzymes, including CYP3A4, CYP2C9, CYP1A2, and CYP2D6, with CYP3A4 being particularly significant in the biotransformation. The standard and extended-release formulations are converted into inactive metabolites, primarily excreted through renal pathways.

Elimination: The controlled-release formulation has a mean elimination half-life of 2.8 hours (range: 1.62 to 4.05 hr). Immediate-release tablets, at 5 mg (range: 1.4 to 4.5 hours) or 10 mg (range: 1.4 to 3.8 hours), exhibit mean elimination half-lives within this specified range. The peak serum concentration (Cmax) of sublingually administered zolpidem is typically achieved within 35 to 75 minutes, and the drug has a half-life of approximately 2.5 hours.

Administration

Available Dosage Forms and Strengths

Zolpidem oral formulation is available as a tablet and an extended-release tablet. Zolpidem is also available as an oral spray formulation sprayed into the mouth over the tongue and a sublingual tablet to place under the tongue. Depending on the patient's sleep quality, it is administered in 5 mg or 10 mg tablets orally. Zolpidem is then converted to an inactive metabolite and excreted by the kidneys. Tablets are not scored. Ingestion with or immediately after food intake may slow the effects of this drug.

Adult Dosing

The recommended initial dose is 5 mg for women and 5 mg or 10 mg for men, taken only once per night immediately before bedtime, with at least 7 to 8 hours remaining before the planned time of awakening. Zolpidem clearance is lower in women.

Specific Patient Populations

Hepatic impairment: The dosage should be changed in patients with hepatic impairment, as zolpidem's half-life was many times longer than in patients with normal liver function.

Renal impairment: Zolpidem pharmacokinetics remain consistent in renal impairment, requiring no dosage adjustment. In patients undergoing hemodialysis for end-stage renal failure, there is no significant difference in key pharmacokinetic parameters for zolpidem between the initial and final administration days. This suggests that renal function does not notably impact zolpidem metabolism or clearance.

Pregnancy considerations: As zolpidem can cross the placenta, maternal use of zolpidem in the third trimester of pregnancy or during labor could lead to respiratory depression and sedation in neonates upon birth. Therefore, it is recommended to monitor and manage neonates exposed to

maternal zolpidem or during labor for signs of hypotonia, excessive sedation, and respiratory depression. However, published data from post-marketing surveillance, observational studies, case reports, and birth registries on the maternal use of zolpidem during pregnancy have not concluded a clear association between zolpidem use and significant congenital disabilities. There are limited moderate to severe cases of respiratory depression reported after birth in neonates who require intratracheal intubation or artificial ventilation. However, once treated, most of these neonates recovered within hours to a few weeks after birth.

Breastfeeding considerations: Limited published literature reported that zolpidem is excreted in human milk and lacks information on its effects on milk production. There are reports of excessive sedation in infants exposed to zolpidem through breast milk. The health and developmental benefits of breastfeeding, the mother's clinical requirement for zolpidem, and any potential adverse reactions on the breastfed infant from zolpidem or the underlying maternal condition should be considered. Infants exposed to zolpidem through breast milk should be monitored for hypotonia, excess sedation, and respiratory depression. To minimize exposure to a breastfed infant, a lactating woman may temporarily interrupt breastfeeding and discard breast milk during treatment and for 23 hours (approximately 5 elimination half-lives) after zolpidem intake.

Pediatric patients: Pediatric patients should not be given zolpidem as its effectiveness has not been found yet. In addition, the research found that hallucinations occur in a small percentage of pediatric patients who received zolpidem.

Older patients: According to the American Geriatrics Society (AGS) 2023 Beers Criteria, nonbenzodiazepine receptor agonist hypnotics like zolpidem and benzodiazepines (BZDs) increase the risk of delirium, fractures, emergency room visits, motor vehicle accidents, and hospitalizations. Clinicians should avoid their use if possible. According to FDA-approved product labeling, the dose should be reduced to 5 mg as serum concentrations are typically higher than in young adults.

Adverse Effects

Adverse effects of zolpidem include anaphylaxis, changes in behavior, withdrawal, and central nervous system (CNS) depression. In rare situations, patients have reported tongue, larynx, or glottis swelling in the form of angioedema. Also, patients have reported shortness of breath, airway closure, nausea, and vomiting. If these events are reported, do not use zolpidem. Patients who experience closure in the throat, glottis, or larynx should be sent to the emergency department.

Drug-Drug Interactions

Alcohol and CNS depressants: Concurrent use of CNS depressants or alcohol with zolpidem may result in additive impairment of psychomotor performance and an increased risk of CNS depression. Due to potential additive effects, adjustments in the dosage of zolpidem and other

concomitant CNS depressants may be necessary. For the 5 mg dose of zolpidem immediate-release, consider employing an alternative immediate-release product. Avoid combining zolpidem with other sedative-hypnotics at bedtime or during the night.

Opioids: Concurrent administration of zolpidem and opioids may elevate the risk of respiratory depression. Restrict the dosage and duration of simultaneous use of zolpidem and opioids; monitor for respiratory depression.

Imipramine and chlorpromazine: Concurrent use of imipramine or chlorpromazine with zolpidem may produce an additive effect, decreasing alertness and psychomotor performance. In patients who are prescribed opioids, concurrent use of Z-drugs (eg, zolpidem) significantly raises the risk of overdose, even after accounting for confounding factors. This finding is particularly significant given the widespread use of Z-drugs among patients treated with opioids.

Sertraline: Concurrent administration of zolpidem and sertraline increases exposure to zolpidem. Consider using another immediate-release product with a lower zolpidem dose (5 mg).

CYP3A4 inducers: Simultaneous zolpidem use with CYP3A4 inducers like rifampin may decrease zolpidem exposure, impacting its metabolism. St. John's Wort induces hepatic CYP enzymes, including CYP3A4 and CYP2B6. This accelerates the metabolism of zolpidem, potentially reducing its effectiveness.

CYP3A4 inhibitors: Simultaneous use of CYP3A4 inhibitors increases zolpidem exposure. Consider using another immediate-release product with a lower zolpidem dose. Additionally, clinicians should be aware that ciprofloxacin (a strong inhibitor of CYP1A2, a moderate inhibitor of CYP3A4) and fluvoxamine (a strong inhibitor of CYP1A2, a weak inhibitor of CYP2C9 and CYP3A4) can inhibit the metabolism of zolpidem, increasing the exposure to zolpidem.

Contraindications

Zolpidem is contraindicated in patients with a known allergy to the drug or inactive ingredients in the formulation. According to updated labeling, zolpidem is also contraindicated in patients with a history of complex sleep behaviors following its administration.

Box Warnings

Complex sleep behaviors can occur after using zolpidem, such as sleep-driving, sleep-walking, and engaging in activities while not fully awake. These actions can lead to serious harm to the individual or others and, in some cases, be fatal. Additionally, other complex behaviors like making phone calls, engaging in sexual activity, or preparing and consuming food have been reported. Patients typically have no recollection of these events. Postmarketing data indicates that complex sleep behaviors can manifest with zolpidem tablets alone, even when taken at recommended

doses, regardless of whether alcohol or other central nervous system (CNS) depressants are used concurrently. If a patient/caregiver reports a complex sleep behavior, it is crucial to discontinue zolpidem immediately.

Warnings and Precautions

Changes in behavior and abnormal thinking have been reported after zolpidem administration. In addition, patients have demonstrated aggressiveness and extroversion uncommon for the person's usual behavior. Patients with alcohol or drug toxicities have experienced auditory and visual hallucinations associated with strange behavior and agitation. Consumption of alcohol or any other CNS depressant increased these events as they enhance sedation when combined. In these cases, the drug needs to be discontinued. Patients who are depressed should also not take zolpidem as it worsens depression and suicidal ideations and actions.

Caution should be used in patients taking drugs that affect drug metabolism via cytochrome P450. Consider giving a lower zolpidem dosage, as patients have shown to enhance sedative effects. In addition, patients taking imipramine and chlorpromazine should avoid using zolpidem. When combined, these medications cause decreased alertness and psychomotor performance.

Worsening of depression or suicidal ideation may occur with zolpidem therapy. Therefore, clinicians should prescribe the least amount of tablets possible to avoid intentional zolpidem overdose. Consider the risk of respiratory depression before prescribing zolpidem in patients with compromised respiratory function. Withdrawal symptoms may occur if the zolpidem dose is tapered off rapidly or discontinued. Therefore, it is recommended to taper off slowly to avoid unnecessary withdrawal and adverse reactions in chronic zolpidem users.

Monitoring

The drug elimination half-life for 5 mg of zolpidem was found to be 2.6 hours. Respectively, the elimination for patients given 10 mg of zolpidem is 2.5 hours, ranging between 1.4 to 3.8 hours. Zolpidem demonstrates a linear kinetics pattern when the dose range is between 5 mg and 20 mg. The drug was also found to be mostly bound to protein and remained unchanged in concentration, subsequently extracted through the renal system.

This drug has a high potential for overuse and daily dependence. Patients with a few weeks of drug use have a low behavioral dependency on zolpidem. Patients who used zolpidem in higher single doses or had a history of drug abuse should be monitored carefully when using zolpidem or any other hypnotic. Patients experience anterograde amnesia after drug administration if plasma concentrations are high during the stimulus. This is attributed to either inattention or consolidation of the memory process.

Toxicity

Drug overdose with zolpidem involves CNS depression, cognitive impairments leading to drowsiness or coma, cardiovascular and respiratory depression, and other fatalities. The acute toxicity of zolpidem is less in severity than other short-acting benzodiazepines like triazolam and midazolam. However, in combined intoxication with other CNS depressant drugs, zolpidem can induce coma in patients even with a low concentration. Single-drug poisoning is benign and should not require therapeutic intervention.

In cases of zolpidem overdose, gastric lavage should be considered within the first hour of ingestion to optimize its benefits. This procedure is most appropriate for conscious patients with a functional gag reflex, as it minimizes the risk of aspiration. Patients can also benefit from the administration of flumazenil and intravenous fluids as well. Flumazenil is a known reversal agent for benzodiazepine toxicity. However, it can exacerbate other neurological symptoms, such as convulsions. In the event of drug toxicity, the patient's respiratory function, oxygen saturation, blood pressure, pulse, and other vitals should be monitored.

Enhancing Healthcare Team Outcomes

Managing insomnia and zolpidem use requires an interprofessional team approach of healthcare professionals, including pharmacists, therapists, nurses, and clinicians (MD, DO, NP, PA). Prescribers should consider the possibility of addiction and withdrawal symptoms when starting patients on zolpidem. Without proper management, significant drug-drug interactions may occur. For example, CNS depression is increased with the combination of opiates, benzodiazepines, antidepressants, or alcohol. Because the drug has the potential to cause dependence, it should not be prescribed for long periods, and the team should coordinate monitoring for dependency concerns. In addition, the patient should be counseled by nurses and pharmacists to avoid alcohol and other CNS depressants when taking zolpidem. Nurses can verify dose and therapy duration before zolpidem administration, and pharmacists can perform medication reconciliation to optimize the treatment with zolpidem and enhance patient outcomes. An RCT investigated the impact of direct-to-patient education, with or without pharmacist consultation, on discontinuing Z-drugs (including eszopiclone, zolpidem, and zaleplon) in older adults. Despite recommendations discouraging their use, these medications are frequently prescribed to manage insomnia in this demographic. The study findings revealed that patients receiving education alone or in conjunction with a pharmacist consultation were significantly more likely to discontinue zolpidem than those receiving standard care.

Review Questions

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